Functions

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Why Functions?

The Problem with Copy-Paste

Scenario: You need to calculate the area of a circle at 5 different places in your program.

Without Functions:

- Copy-paste the formula 5 times: $\text{area} = 3.14159 * \text{r} * \text{r};$
- What if the formula is wrong? Fix it in all 5 places!
- What if the requirement changes? Modify all 5 places!
- Code becomes long, repetitive, and error-prone.

With Functions:

- Write the logic once in a function.
- Call the function wherever needed.
- Need to fix/change? Modify only the function!

Principle: DRY - Don’t Repeat Yourself!

What is a Function?

Definition

A **function** is a self-contained block of code that performs a specific task. It can take input (parameters), process it, and optionally return a result.

Analogy: A Vending Machine

- **Input:** You put in money and press a button (parameters)
- **Process:** Machine selects the item (function body)
- **Output:** You get a snack (return value)

Benefits of Functions:

- **Modularity:** Break complex problems into smaller parts
- **Reusability:** Write once, use many times
- **Maintainability:** Easier to debug and update
- **Abstraction:** Hide complex details behind a simple interface

Functions You Already Know

Function	Purpose	Header File
`printf()`	Print formatted output	`<stdio.h>`
`scanf()`	Read formatted input	`<stdio.h>`
`sqrt()`	Square root	`<math.h>`
`pow()`	Power (x^y)	`<math.h>`
`strlen()`	String length	`<string.h>`
`main()`	Program entry point	(Required)

Key Insight: `main()` is itself a function! Every C program must have exactly one `main()` function

Function Syntax and Structure

Anatomy of a Function

General Syntax

```
return_type function_name (parameter_list) {
    // Function body
    // Local variable declarations
    // Statements
    return value; // Optional (required if return_type is not void)
}
```

Components:

1. **Return Type:** What type of value the function gives back (`int`, `float`, `void`, etc.)
2. **Function Name:** Identifier following same rules as variables
3. **Parameter List:** Input values (can be empty)
4. **Function Body:** The actual code enclosed in `{}`
5. **Return Statement:** Value sent back to the caller

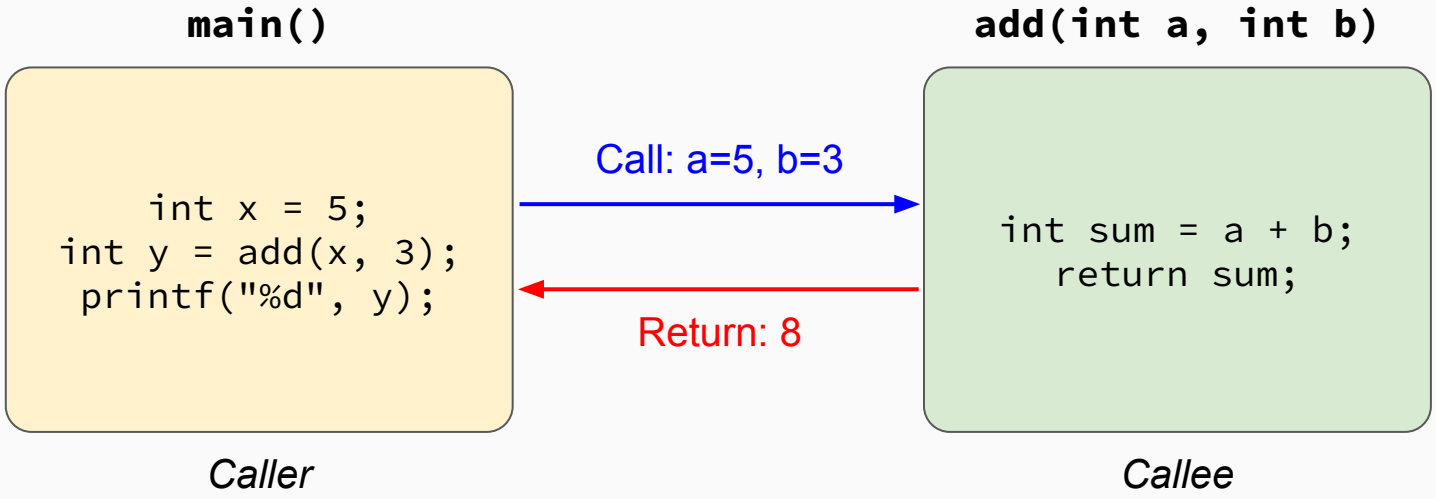
Example: A Simple Function

```c
#include <stdio .h>
// Function definition
int add (int a, int b) {
     int sum = a + b;
     return sum;
}

int main () {
     int result = add(5, 3); // Function call
     printf ("Sum = %d\n", result);
     // Can also use directly
     printf("Another sum = %d\n", add(10, 20));
     return 0;
}
```

Output:

```
Sum = 8
Another sum = 30
```

The Function Calling Process



Flow: `main()` → calls `add()` → `add()` executes → returns to `main()`

Function Declaration vs Definition

Declaration (Prototype):

- Tells the compiler the function exists
- Placed before `main()` or in header files
- Ends with a semicolon

Definition:

- Contains the actual code
- Can be placed anywhere
- No semicolon after)

```
// Declaration only
int add (int a, int b);
// Parameter names optional
int add (int , int);
```

```
// Full definition
int add (int a, int b) {
    return a + b;
}
```

Why Use Prototypes?

Allows you to define functions **after** `main()`, making code more readable (main at top).

Complete Example: Using Prototypes

```c
#include <stdio.h>

// Function prototypes (declarations)
int add (int a, int b);
int multiply (int a, int b);

int main () {
     printf ("5 + 3 = %d\n", add(5, 3));
     printf ("5 * 3 = %d\n", multiply(5, 3));
     return 0;
}

// Function definitions (can be after main)
int add (int a, int b) {
     return a + b;
}
int multiply (int a, int b) {
     return a * b;
}
```

Parameters and Arguments

Parameters vs Arguments

Often confused but important distinction!

Parameters (Formal):

- Variables in function **definition**
- Act as placeholders
- Like variable declarations

```
int add(int a, int b)
```

`a` and `b` are parameters

Arguments (Actual):

- Values passed when **calling**
- The actual data
- Can be literals, variables, expressions

```
result = add(5, x+2);
```

`5` and `x+2` are arguments

Memory Analogy

Parameters are like empty boxes with labels. Arguments are the actual items you put in those boxes when calling the function.

Types of Arguments

Arguments can be:

```
int x = 10, y = 20;
// 1. Literal values
result = add(5, 3);
// 2. Variables
result = add(x, y);
// 3. Expressions
result = add(x + 1, y * 2);
// 4. Other function calls
result = add(add(1, 2), add(3, 4)); // add (3, 7) = 10
// 5. Mixed
result = add(x, 100) ;
```

Key Point: Arguments are evaluated **before** being passed to the function.

Multiple Parameters and Order

```
// Function with multiple parameters of different types
float calculateBMI (float weight, float height) {
     return weight / (height * height);
}

// Order matters!
float bmi1 = calculateBMI(70.0, 1.75); // Correct : weight =70 , height =1.75
float bmi2 = calculateBMI(1.75, 70.0); // WRONG : weight =1.75 , height =70!
```

Critical Rule

Arguments are matched to parameters **by position**, not by name!

- 1st argument → 1st parameter
- 2nd argument → 2nd parameter
- ... and so on

Call by Value vs Call by Reference

Two Ways to Pass Arguments

Call by Value:

- Function receives a copy of the argument
- Changes to parameter don’t affect original
- Original data is protected
- C uses this by default!

Analogy: Giving someone a photocopy of your document. They can modify it, but your original is safe.

Call by Reference:

- Function receives the address of the argument
- Changes to parameter affect the original
- More efficient for large data
- Not directly in C, but can be emulated with pointers

Analogy: Telling someone your home address. They can come and rearrange your furniture!

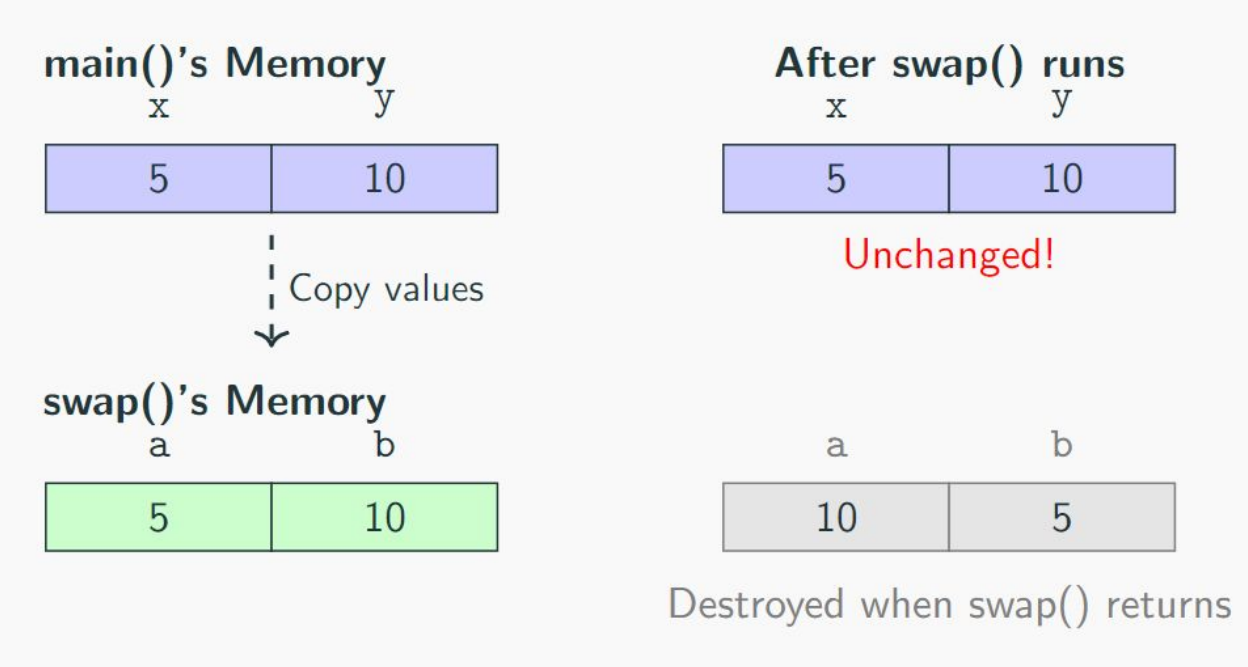
Call by Value: The Swap Problem

```
#include <stdio.h>

void swap (int a, int b) {
      int temp = a;
      a = b;
      b = temp;
      printf("Inside swap: a=%d, b=%d\n", a, b);
}

int main () {
      int x = 5, y = 10;
      printf("Before swap: x=%d, y=%d\n", x, y);
      swap(x, y);
      printf("After swap: x=%d, y=%d\n", x, y);
      return 0;
}
```

Output:

```
Before swap: x=5, y=10
Inside swap: a=10, b=5
After swap:  x=5, y=10 <- x and y unchanged!
```

Memory Visualization: Why Swap Fails



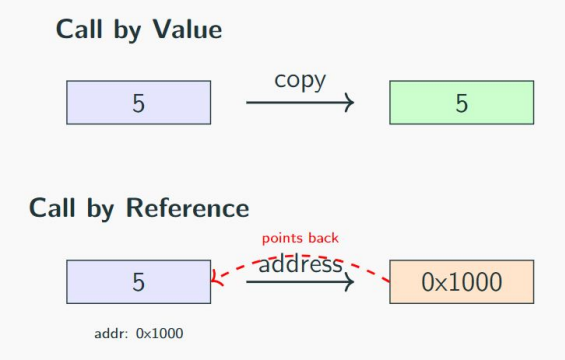
Emulating Call by Reference with Pointers

The Idea: Instead of passing the value, pass the address (location in memory).

Key Concepts (Preview)

Arguments are matched to parameters **by position**, not by name!

- `&x` gives the **address** of variable `x`
- A **pointer** is a variable that stores an address
- `*ptr` accesses the value at the address stored in pointer `ptr`



Swap That Actually Works (Using Pointers)

```c
#include <stdio .h>
void swap (int *a, int *b) { // Receive addresses
      int temp = *a; // Get value at address a
      *a = *b; // Put value at b into address a
      *b = temp ; // Put temp into address b
}
int main () {
      int x = 5, y = 10;
      printf("Before: x=%d, y=%d\n", x, y);
      swap(&x, &y); // Pass addresses!
      printf("After: x=%d, y=%d\n", x, y);
      return 0;
}
```

Output:

Before: `x=5, y=10`
After: `x=10, y=5` <- Actually swapped!

No need to learn pointers in detail. For now, just understand the concept.

Why scanf() Uses &

Now you understand why `scanf()` requires the `&` operator!

```
int age;
scanf("%d", &age);
```

Explanation:

- scanf() needs to modify the variable age
- If we passed age by value, scanf() would only get a copy
- The copy would be modified, but age would remain unchanged!
- By passing &age, we give scanf() the address where to store the input

Remember

Any function that needs to modify its argument must receive the **address**, not the value. This is why C uses call by value by default – it's safer!

Return Types and the return Statement

Understanding Return Values

The return Statement

- Sends a value back to the calling function
- Immediately terminates the function execution
- The returned value replaces the function call in the expression

Return Types in C:

Type	Description
`int`	Returns an integer
`float / double`	Returns a decimal number
`char`	Returns a character
`void`	Returns nothing

Rule: The return type in the function definition must match the declared return type.

The void Return Type

Use `void` when a function performs an action but doesn't need to return a value.

```
void greet (char name []) {
     printf ("Hello, %s!\n", name);
     // No return statement needed (or use: return;)
}

void printLine () {
     printf("================\n");
}

int main () {
     greet("Alice");
     printLine();
     // ERROR : Can't assign void to a variable
     // int x = greet("Bob"); <- Compilation error!
     return 0;
}
```

Multiple Return Statements

A function can have multiple return statements, but only one will execute.

```
int absolute(int n) {
     if (n >= 0) {
          return n; // Exit point 1
     } else {
          return -n; // Exit point 2
     }
     // Code here is UNREACHABLE
}

char getGrade(int score) {
     if (score >= 90) return 'A';
     if (score >= 80) return 'B';
     if (score >= 70) return 'C';
     if (score >= 60) return 'D';
     return 'F'; // Default case
}
```

Return Type Mismatch

Question: What happens here?

```
int divide(int a, int b) { return a / b; }
float divideProperly(int a, int b) { return a / b; }

int main() {
     printf("divide(7, 2) = %d\n", divide(7, 2));
     printf("divideProperly(7, 2) = %f\n", divideProperly(7, 2));
     return 0;
}
```

Output:

```
divide(7, 2) = 3
divideProperly(7, 2) = 3.000000 <- Still 3, not 3.5!
```

Why? `a / b` performs integer division first ($7/2 = 3$), then converts to float.

Fix: `return (float)a / b;` or `return a / (float)b;`

Variable Scope and Lifetime

What is Scope?

Definition

Scope refers to the region of the program where a variable is visible and accessible.

Types of Scope in C:

1. **Local Scope:** Variables declared inside a function or block
2. **Global Scope:** Variables declared outside all functions
3. **Block Scope:** Variables declared inside {} blocks

Lifetime:

- **Local variables:** Created when function is called, destroyed when it returns
- **Global variables:** Exist for the entire program duration

Local vs Global Variables

```c
#include <stdio.h>

int globalCount = 0; // Global: visible everywhere
void increment() {
      int localVar = 5; // Local : only in this function
      globalCount++; // Can access global
      printf("Local: %d, Global: %d\n", localVar, globalCount);
}
int main() {
      increment(); // Local: 5, Global: 1
      increment(); // Local: 5, Global: 2
      // printf("%d", localVar); // ERROR: localVar not visible
      printf("Final global: %d\n", globalCount); // 2
      return 0;
}
```

Warning

Global variables can be modified from anywhere! Use sparingly.

Shadowing: When Names Collide

Question: What is the output?

```
#include <stdio.h>
int x = 100; // Global x
void func() {
      int x = 50; // Local x shadows global x
      printf ("In func: x = %d\n", x);
}
int main() {
      int x = 25; // Local x in main
      printf("In main (before): x = %d\n", x);
      func();
      printf("In main (after): x = %d\n", x);
      return 0;
}
```

Output:

In main (before): `x = 25`
In func: `x = 50`
In main (after): `x = 25`

Shadowing: When Names Collide

What if you want a local variable to **remember** its value between function calls?

```
int counter() {
     static int count = 0; // Initialized only ONCE at program start
     count++;
     return count ;
}
int main() {
     printf("%d\n", counter()); // 1
     printf("%d\n", counter()); // 2
     printf("%d\n", counter()); // 3
     return 0;
}
```

Key Points:

- static variables are initialized only once
- They retain their value between function calls
- Useful for counters, caching, maintaining state

Practical Function Examples

Problem: Prime Number Checker

Task

Write a function that takes an integer as parameter and returns 1 if prime, 0 otherwise. Use it to print all primes from 2 to n.

Let’s break this down:

1. First, understand what a prime number is
2. Design the isPrime() function
3. Optimize the function
4. Use the function to solve the complete problem

Step 1: Understanding Prime Numbers

Definition

A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.

Examples:

- Prime: $2, 3, 5, 7, 11, 13, 17, 19, 23, \ldots$
- Not Prime: 1 (by definition), $4\ (2\times2)$, $6\ (2\times3)$, $9\ (3\times3)$, ...

Key Observations:

- 2 is the only even prime number
- To check if n is prime, we need to verify no number from 2 to n − 1 divides n
- If n has a divisor d where $d > \sqrt{n}$, then $n/d < \sqrt{n}$ (so we only need to check up to $\sqrt{n}$)

Step 2: Naive isPrime() Function

Approach: Check if any number from 2 to n-1 divides n.

```
int isPrime(int n) {
     // Handle edge cases
     if (n <= 1) return 0; // 0 and 1 are not prime
     // Check all numbers from 2 to n -1
     for (int i = 2; i < n; i++) {
          if (n % i == 0) {
               return 0; // Found a divisor , not prime
          }
     }
     return 1; // No divisor found , it 's prime
}
```

Analysis:

- For `isPrime(100)`, we check: 2, 3, 4, 5, ..., 99 (98 checks!)
- For `isPrime(1000000)`, we do 999,998 checks!
- Can we do better?

Step 3: Optimization 1 – Check Only Up to $\sqrt{n}$

Key Insight: If $n = a \times b$ and $a \leq b$, then $a \leq \sqrt{n}$.

```
int isPrime(int n) {
     if (n <= 1) return 0;
     // Only check up to sqrt (n)
     for (int i = 2; i * i <= n; i++) { // i * i <= n is same as i <= sqrt (n)
          if (n % i == 0) {
               return 0;
          }
     }
     return 1;
}
```

Improvement:

- For `isPrime(100)`, we check: 2, 3, ..., 10 (only 9 checks!)
- For `isPrime(1000000)`, we check up to 1000 (only 999 checks!)

Step 3: Optimization 2 – Skip Even Numbers

Key Insight: 2 is the only even prime. All other even numbers are not prime.

```
int isPrime(int n) {
     if (n <= 1) return 0;
     if (n > 2 && n % 2 == 0) return 0;
     // Only check up to sqrt (n)
     for (int i = 3; i * i <= n; i+=2) { // i * i <= n is same as i <= sqrt(n)
          if (n % i == 0) {
               return 0;
          }
     }
     return 1;
}
```

Further Improvement:

- We now check only half the numbers: 3, 5, 7, 9, ...
- Combined with $\sqrt{n}$ limit, this is very efficient!

Step 4: The Complete Program

```
#include <stdio.h>
int isPrime(int n) {
      if (n <= 1) return 0;
      if (n > 2 && n % 2 == 0) return 0;
      for (int i = 3; i * i <= n; i += 2) {
            if(n % i == 0) return 0;
      }
      return 1;
}
int main() {
      int n;
      printf("Enter n: ");
      scanf("%d", &n);
      printf("Prime numbers from 2 to %d:\n", n);
      for (int i = 2; i <= n; i ++) {
            if (isPrime(i)) printf ("%d ", i);
      }
      printf("\n");
      return 0;
}
```

Example: GCD (Greatest Common Divisor)

Using the Euclidean algorithm: `gcd(a, b) = gcd(b, a mod b)` until `b = 0`

```
int gcd(int a, int b) {
     if (a < 0) a = -a; // Handle negatives
     if (b < 0) b = -b;

     while (b != 0) {
          int temp = b;
          b = a % b;
          a = temp;
     }
     return a;
}

int main() {
     printf("GCD(48, 18) = %d\n", gcd(48, 18) ); // 6
     printf("GCD(54, 24) = %d\n", gcd(54, 24) ); // 6
     return 0;
}
```

Example: Reverse an Integer

Reverse the digits of a number (e.g., 1234 → 4321)

```
int reverseNumber(int n) {
     int reversed = 0;
     int negative = (n < 0);
     if (negative) n = -n;
     while (n > 0) {
          int digit = n % 10;
          reversed = reversed * 10 + digit;
          n = n / 10;
     }
     return negative ? -reversed : reversed ;
}

int main() {
     printf("Reverse of 1234 = %d\n", reverseNumber(1234)); // 4321
     printf("Reverse of -567 = %d\n", reverseNumber(-567) ); // -765
     return 0;
}
```

Example: Count Digits

Count how many digits are in a number

```
int countDigits(int n) {
     if (n == 0) return 1; // Special case : 0 has 1 digit
     if (n < 0) n = -n; // Handle negative
     int count = 0;
     while (n > 0) {
          count++;
          n = n / 10;
     }
     return count;
}

int main() {
     printf("Digits in 12345 = %d\n", countDigits(12345)); // 5
     printf("Digits in 0 = %d\n", countDigits(0)); // 1
     printf("Digits in -99 = %d\n", countDigits(-99)); // 2
     return 0;
}
```

Header Files in C

What are Header Files?

Definition

Header files (.h files) contain function declarations, macros, and type definitions that can be shared across multiple source files.

Why Use Header Files?

- **Code Organization:** Separate interface from implementation
- **Reusability:** Share functions across multiple programs
- **Modularity:** Break large programs into manageable pieces
- **Standard Library:** Access pre-written functions

Two Types:

- `<header.h>` – System/standard headers (searched in system directories)
- `"header.h"` – User-defined headers (searched in current directory first)

Common Standard Header Files

Header	Common Functions
`<stdio.h>`	`printf, scanf, fopen, fclose`
`<stdlib.h>`	`malloc, free, atoi, rand, exit`
`<string.h>`	`strlen, strcpy, strcmp, strcat`
`<math.h>`	`sqrt, pow, sin, cos, log`
`<ctype.h>`	`isalpha, isdigit, toupper, tolower`
`<time.h>`	`time, clock, difftime`
`<stdbool.h>`	`bool, true, false (C99)`

Note for `<math.h>`

When using math functions, compile with `-lm` flag: `gcc program.c -lm`

Creating Your Own Header File

Step 1: Create the header file `mymath.h`

```
// mymath.h - Function declarations (prototypes)
#ifndef MYMATH_H // Include guard: prevents double inclusion
#define MYMATH_H
int add(int a, int b);
int multiply(int a, int b);
int factorial(int n);
#endif
```

Step 2: Create the implementation file `mymath.c`

```
// mymath.c - Function definitions
#include "mymath.h"
int add(int a, int b) { return a + b; }
int multiply(int a, int b) { return a * b; }
int factorial(int n) {
     if (n <= 1) return 1;
     return n * factorial (n - 1);
}
```

Using Your Custom Header

Step 3: Use in your main program main.c

```
// main.c
#include <stdio.h>
#include "mymath.h" // Include YOUR header with quotes
int main () {
      printf ("5 + 3 = %d\n", add(5, 3));
      printf ("5 * 3 = %d\n", multiply(5, 3));
      printf ("5! = %d\n", factorial(5));
      return 0;
}
```

Step 4: Compile all files together

```
gcc main .c mymath .c -o program
./ program
```

The Flow: `main.c` includes `mymath.h` (declarations) → Linker connects to `mymath.c` (definitions)

Summary: Key Takeaways

1. **Functions** help organize code, enable reuse, and simplify maintenance
2. **Parameters** are placeholders; **arguments** are actual values passed
3. C uses **call by value** – functions get copies of arguments
4. **Call by reference** can be emulated using pointers (addresses)
5. The **return statement** sends a value back and exits the function
6. **Scope**: Variables are only visible in their declared region
7. **Header files** organize and share code across multiple files

Golden Rules

- One function = One task (Single Responsibility)
- Keep functions short and focused
- Use meaningful names for functions and parameters

Acknowledgement

- Mohammad Sadat Hossain, Lecturer, CSE, BUET